a a b c

i/p: a a b c
o/p: a # b b

a a b c
a # b b

⇒ [q: ~~a~~ b c].

freq.
a → ~~1~~ 2
b → 1
[c → 1]

[-8, 2, 3, -6, 10]   k = 2.

[-8, 0, -6, -6]

[12, -1, -7, 8, -15, 30, 16, 28]   k=3.

[-1, -1, -7, -15, -15, 0]

BF: TC: O(nk)
      AS: O(1).

0 1 2 3 4 5 6 7   n-k
[12, -1, -7, 8, -15, 30, 16, -28].   k=3.

q: ~~1~~, ~~2~~, ~~4~~, 7.

~~-1~~ ~~-7~~

~~-7~~, ~~-1~~, ~~-7~~

res: -1, -1, -7, -15, -15, -28

0 1 2 3 4
[-8, 2, 3, -6, 10)   k = 2.

q: ~~2~~ 3

res: -8, 0, -6, -6.

AS: O(k).

[1, 2, 3, 1, 4, 5, 2, 3, 6]   k=3

[3, 3, 4, 5, 5, 5, 6]

BF: TC: O(nk).
      AS: O(1).

0 1 2 3 4 5 6 7 8
[1 2 3 1, 4, 5, 2, 3, 6]   k=3.

~~1~~ ~~2~~ ~~3~~, ~~1~~, ~~4~~, ~~5~~, ~~2~~ ~~3~~ 6

res: 3, 3, 4, 5, 5, 5, 6

...5...4...6.

...5...4 - 2...3

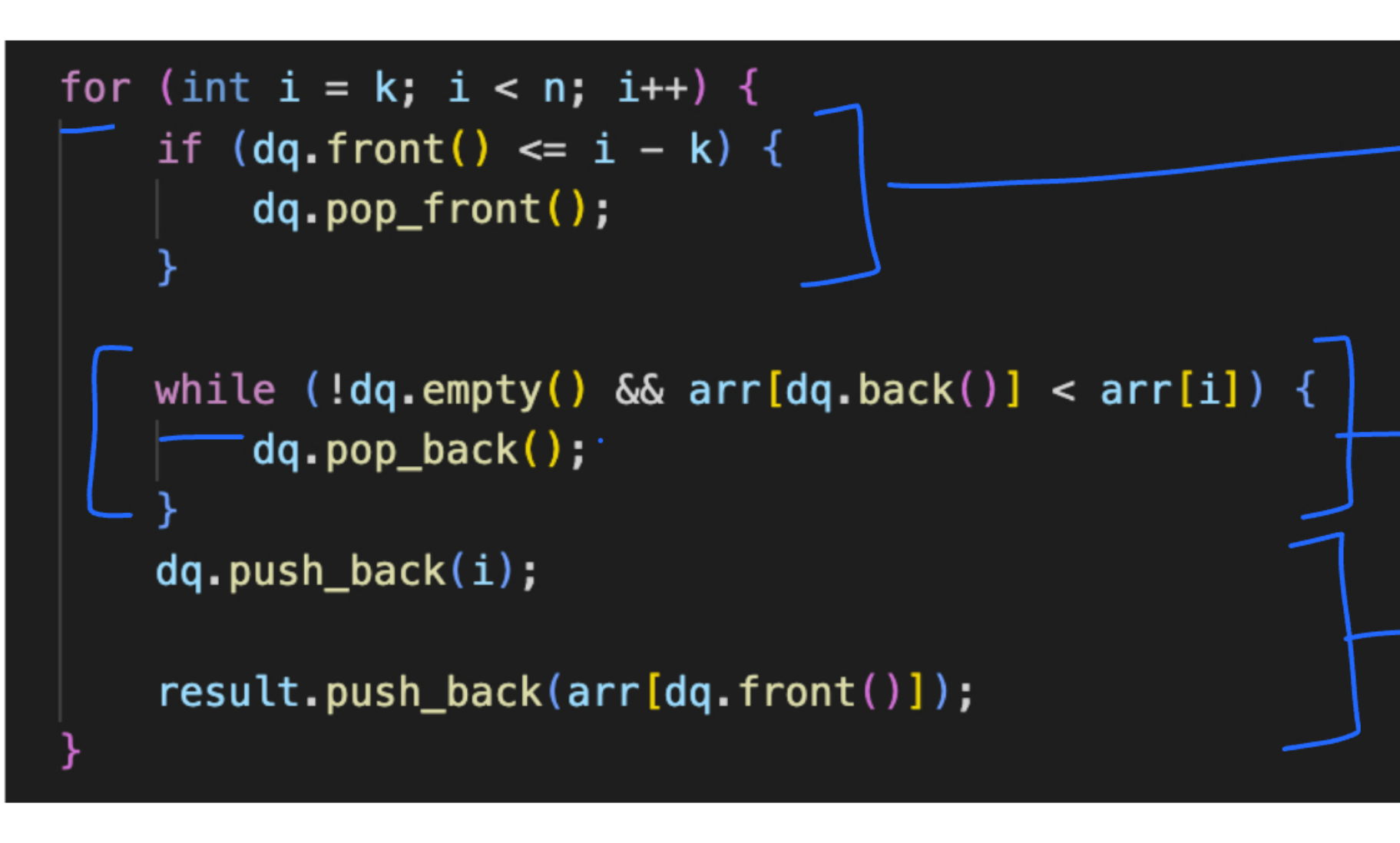
~~5~~, ~~4~~ 6.

5, 4, ~~2~~, 3

Deque.

- push_back() — Q
- pop_back() — S
- back() — S
- front() — Q
- pop_front(). — Q

AS: O(k).
TC:



```
for (int i = k; i < n; i++) {
    if (dq.front() <= i - k) {
        dq.pop_front();
    }

    while (!dq.empty() && arr[dq.back()] < arr[i]) {
        dq.pop_back();
    }
    dq.push_back(i);

    result.push_back(arr[dq.front()]);
}
```

→ O(n).
→ O(n).
→ O(n).

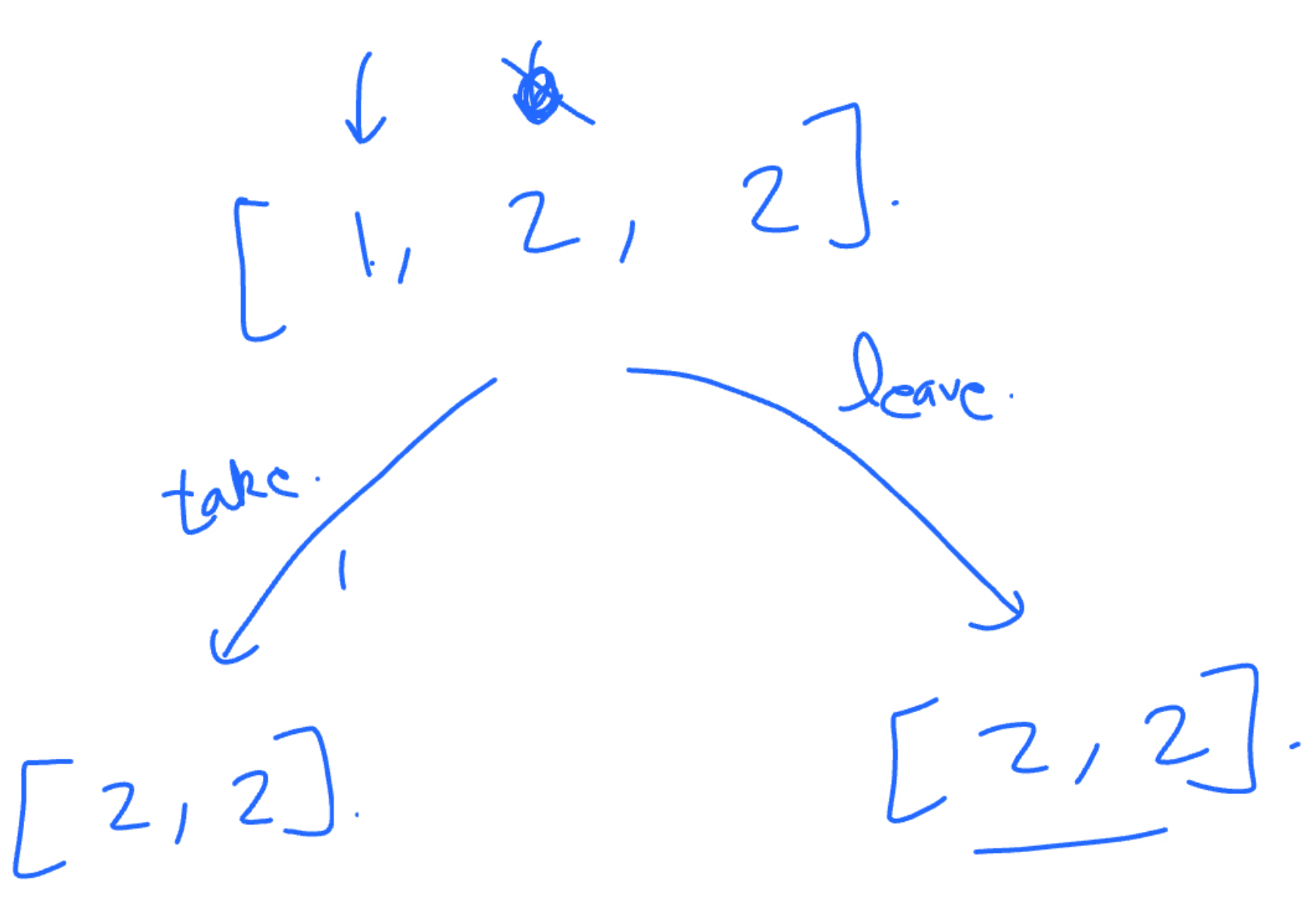
⇒ TC: O(n).

Unique. Subsets.

[2, 1, 2]

{} {1} {2} {1, 2} {2, 2} {1, 2, 2}

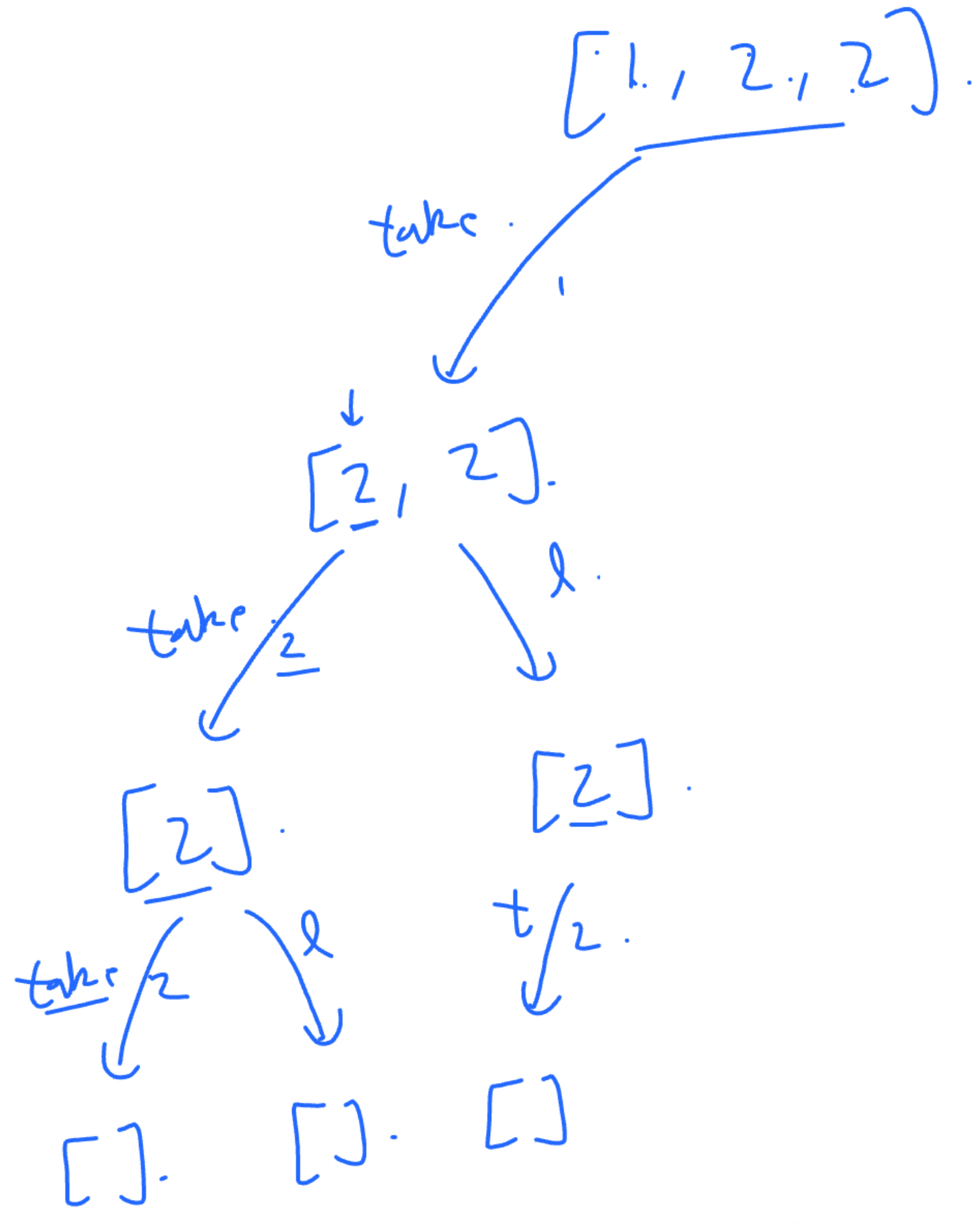
~~{2}~~   ~~{2}~~

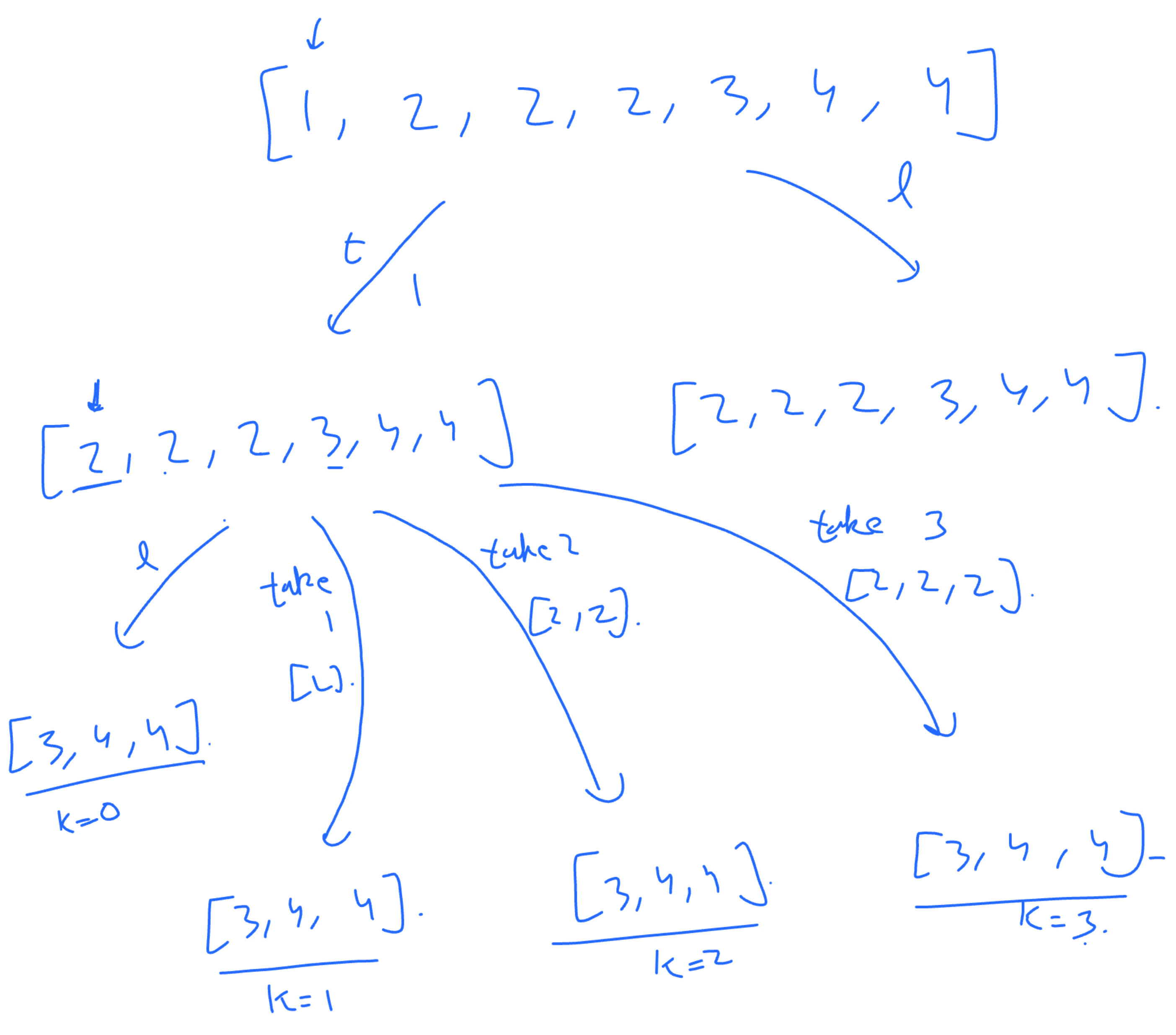

[1, 2, 2].
take.
1
leave.
[2, 2].
[2, 2].

[~~1~~ ~~x~~ 2]

ttl.
tlt

curr: 1, ~~2~~, ~~2~~ 2
res: [1, 2, 2].
→ [1, 2].
[1, 2].


[1, 2, 2].
take.
1
[2, 2].
take 2
l.
[2].
[2].
take 2
l
t 2.
[].
[].
[]


0 1 2 3 4 5 6
[1, 2, 2, 2, 3, 4, 4]
t
1
l
[2, 2, 2, 3, 4, 4]
[2, 2, 2, 3, 4, 4].
l
take 1
[2].
take 2
[2, 2].
take 3
[2, 2, 2].
[3, 4, 4].
k=0
[3, 4, 4].
k=1
[3, 4, 4].
k=2
[3, 4, 4].
k=3.